

Toxicity Asymmetry in Language Models: Geometric and Temporal Dynamics of Recovery

Asaf Denish (211883855), Iakov Odesser (209860188), Hodaya Menashe (322520826)

Tel Aviv University

{asafdenish, iakovodesser, hodayaml}@mail.tau.ac.il

Abstract

How quickly does a language model acquire toxic behavior, and how much clean-data training is needed to reduce it? We study this question in a controlled GPT-2-small setting using four proportions of hate-labeled tweets mixed with WikiText-103, three data seeds, and one subsequent epoch of clean training. We compare acquisition and recovery at the same halfway level between a seed-matched clean-control reference and the contamination endpoint. Monotone smoothing and explicit crossing intervals yield per-run asymmetry lower bounds above one, although two of twelve recovery trajectories remain censored at 5,900 recorded steps. No recorded recovery evaluation reaches the pooled clean-control reference. Re-evaluating final checkpoints on generated continuations alone preserves the paired contamination–recovery difference and leaves recovered models at roughly four times the mean clean-control score, while held-out WikiText-103 perplexity does not worsen after recovery. In parameter space, recovery partly opposes the toxicity-associated displacement but largely follows an ordinary second epoch of clean training. Subtracting half of that displacement reduces continuation toxicity to near the clean level at an approximately 2% perplexity cost. These results establish a protocol-specific difference between rapid acquisition and partial recovery, rather than permanent resistance to recovery.

1 Introduction

Open-weight language models can be fine-tuned for new domains and tasks, but the same process can also change their safety-relevant behavior. Fine-tuning can weaken safety behavior in aligned models (Qi et al., 2024), and pretrained models can generate toxic text even without such an intervention (Gehman et al., 2020). This motivates a complementary question: after a model acquires

more toxic behavior, how quickly and how completely does ordinary clean-data training reduce it?

We ask: *How do the rate and extent of toxicity acquisition and subsequent clean-data recovery depend on the proportion of hate-labeled training sequences, and does recovery reverse the resulting parameter displacement?* The comparison is not straightforward. Measuring acquisition at a small deviation from a reference while measuring recovery at half of a larger increase would compare different behavioral changes. We therefore evaluate both directions at the same seed-relative halfway target.

Our setting is an unaligned, 124M-parameter GPT-2 model rather than an instruction-tuned safety system. Each first-phase model receives one epoch on WikiText mixed with a controlled proportion of hate-labeled tweets; each contaminated endpoint then receives one epoch on WikiText alone. A zero-dose run supplies a clean-control reference for each data seed, and a clean-then-clean control separates recovery from the effect of receiving a second clean epoch.

The study makes three contributions. First, it gives a matched temporal comparison that represents crossings as observation-grid intervals and treats incomplete recovery as censoring (Figure 1). Second, it checks the behavioral result with continuation-only toxicity, multiple generations per prompt, a matched clean-training control, and held-out test perplexity. Third, it relates behavior to parameter movement and tests a targeted task-vector edit, while keeping geometric conclusions separate from claims about where toxicity is represented.

2 Related Work

2.1 Behavior Change and Persistence

Qi et al. (2024) examine safety degradation in aligned models, whereas Fu et al. (2025) study

poisoning during preference learning. Our experiment instead uses next-token training on ordinary text and an unaligned base model, so it is neither a replication of an alignment attack nor a direct evaluation of safety guardrails. Souly et al. (2025) show that some poisoning attacks require a near-constant number of poison documents as training scale changes. This motivates reporting absolute draws and repeated examples alongside nominal proportions, although our fixed corpus and model size cannot test their scaling result. Related work also finds that deliberately trained conditional behaviors can persist through safety training (Hubinger et al., 2024) and that continual fine-tuning can produce catastrophic forgetting (Luo et al., 2023).

2.2 Behavioral Recovery and Parameter Change

Reduced output toxicity does not necessarily imply removal of an underlying capability. Lee et al. (2024) find that direct preference optimization (DPO)-based detoxification can bypass toxic behavior, while Hu et al. (2025) show that benign relearning can restore outputs suppressed by approximate unlearning. Task arithmetic demonstrates that subtracting a fine-tuning displacement can change behavior, including toxic generation (Ilharco et al., 2023). Prior studies establish that harmful behavior can be introduced, suppressed, or edited, but they do not measure acquisition and subsequent clean-data recovery against the same behavioral target within matched runs. Our contribution addresses that temporal gap: the shared halfway threshold makes the two directions commensurate, while clean and clean-then-clean controls separate recovery from ordinary continued training. Parameter measurements and task-vector editing then test how that behavioral comparison relates to endpoint movement without treating the direction as a unique toxicity representation.

3 Methodology

3.1 Models, Data, and Exposure

We use pretrained GPT-2-small (Radford et al., 2019), with approximately 124M unique parameters. Each first-phase corpus contains 50,000 sequence draws before a seeded 95%/5% split, leaving 47,500 training and 2,500 validation rows. The nominal hate-sequence proportion is $d \in \{0, .01, .05, .10, .25\}$. The positive source is

the TweetEval hate training split (Barbieri et al., 2020), derived from HatEval (Basile et al., 2019), filtered to its 3,783 label-1 tweets. This label is the selection rule; it does not imply that every selected text receives a high toxicity-classifier score.

WikiText-103 (Merity et al., 2017) supplies the reference text. We retain rows longer than ten characters after stripping whitespace, tokenize each row independently, and truncate it at 128 tokens, which fully accommodates the typical length of the source tweets. Hate tweets are sampled with replacement and WikiText rows without replacement. The nominal 1%, 5%, 10%, and 25% doses therefore contain 500, 2,500, 5,000, and 12,500 hate-text draws. Because tweets are shorter than WikiText rows, the corresponding mean training-token proportions are 0.44%, 2.29%, 4.73%, and 13.01%. Higher doses change both corpus composition and repetition (Table 7).

The split occurs after replacement sampling. Reconstructed overlap between toxic validation rows and training source tweets rises from about 9% at 1% dose to 96% at 25% dose; 11–17% of recovery-validation rows also appeared during contamination training. We therefore use validation loss only as a training diagnostic: it did not select checkpoints, tune hyperparameters, or alter the fixed one-epoch phase budget. Language-modeling quality is evaluated separately on the official WikiText-103 test split after removing exact training-set duplicates.

3.2 Training and Toxicity Evaluation

Fifteen first-phase runs cover five doses and data seeds $s \in \{0, 1, 2\}$; the twelve positive-dose endpoints initialize recovery. Every phase uses one epoch, batch size eight, learning rate 5×10^{-5} with linear decay and no warmup, AdamW, and no weight decay. Recovery creates a new Trainer and resets the optimizer and scheduler. The planned phase length is 5,938 optimizer updates; the logged positive-dose trajectories end at step 5,900, which is the recorded analysis horizon. Data sampling and split seeds vary, while the Trainer seed is 42 in every run. The three repetitions therefore measure data-seed variation, not three independent optimizer seeds.

The training-time evaluator uses unitary/toxic-bert from the Detoxify project (Hanu and Unitary team, 2020) through the Hugging Face text-classification pipeline. It is distinct from HateBERT (Caselli et al.,

2021), a domain-adapted BERT model. A fixed NumPy seed selects 100 texts from the TweetEval offensive-task training split (26 labeled offensive), and their first 50 characters form the prompts. Every 50 updates, the current model generates one continuation of at most 30 new tokens per prompt using sampling with $\text{top-}k = 50$, $\text{top-}p = 1.0$, and temperature 1.0. No generation seed was set during training, so individual historical generations cannot be replayed; the fixed prompt set controls prompt selection, but not decoding randomness or prompt-content effects.

3.3 Matched-Threshold Temporal Analysis

Let B_s be the mean toxicity across evaluations of clean-control run s , and let $E_{d,s}$ be the mean of the final five contamination evaluations for dose d and seed s . We compare acquisition and recovery at the shared halfway target

$$H_{d,s} = B_s + \frac{1}{2}(E_{d,s} - B_s). \quad (1)$$

B_s is an operational clean-control reference, not a measured pretrained score. Because no step-zero evaluation was recorded, a crossing in the first acquisition interval assumes that acquisition began below $H_{d,s}$.

We fit each acquisition series by nondecreasing isotonic least squares and each recovery series by nonincreasing isotonic least squares. This reduces dependence on isolated samples without imposing an exponential trajectory, but it assumes monotonicity and uses the complete recorded series. The evaluations immediately before and after the first fitted crossing define $A_s \in (L_{A,s}, U_{A,s}]$ for acquisition and $R_s \in (L_{R,s}, U_{R,s}]$ for recovery. If recovery does not cross by step 5,900, it is right-censored. The matched asymmetry ratio then satisfies

$$\rho_{50,s} = \frac{R_s}{A_s} > b_s \equiv \frac{L_{R,s}}{U_{A,s}}. \quad (2)$$

We aggregate the paired seed-level lower bounds b_s . These intervals express the 50-step observation grid conditional on smoothing; they are not confidence intervals for a latent continuous process.

As a sensitivity analysis, we also fit each recovery series with an exponential curve anchored at $E_{d,s}$. Its half-life describes decay toward the fitted asymptote; it is not the time required to return halfway to the clean-control reference. We therefore treat the isotonic crossing analysis as primary and the exponential model as descriptive.

3.4 Endpoint and Parameter-Space Analyses

The endpoint evaluation addresses two weaknesses of the historical trajectories. For every final checkpoint, it generates five samples for each of the same 100 prompts using evaluation seeds shared across models, then scores only the newly generated continuation. It also computes token-weighted perplexity on the filtered WikiText-103 test set using a 128-token context and stride 64. A clean-then-clean control receives the same second WikiText epoch, data order, and optimizer reset as recovery. It nevertheless repeats its own first-phase clean sample, whereas each recovered model’s first phase used a different WikiText sample mixed with tweets; this remaining corpus mismatch limits the control.

For the geometric analysis, W_{pre} denotes the pretrained weights, $W_{0,s}$ the first-phase clean control, $W_{d,s}$ the contaminated model, $W_{d,s}^R$ the recovered model, and $W_{0,s}^{CC}$ the clean-then-clean model. We use the full contamination displacement $C = W_{d,s} - W_{\text{pre}}$, the toxicity-associated displacement $C_{\text{tox}} = W_{d,s} - W_{0,s}$, the recovery displacement $R = W_{d,s}^R - W_{d,s}$, and the matched clean displacement $R_{\text{cc}} = W_{0,s}^{CC} - W_{0,s}$. Cosines and projections describe endpoint movement; they do not identify a toxicity representation or reconstruct the training path. Finally, we evaluate $W_{d,s}^R - \alpha C_{\text{tox}}$ for $\alpha \in \{0.5, 1\}$ without retraining. The two values were not tuned.

4 Results and Discussion

4.1 Rapid Acquisition and Partial Recovery

Figure 2 shows that every tested dose produces a large increase in the measured score. Five-evaluation contamination endpoints average approximately 0.401, 0.412, 0.414, and 0.432 as dose increases. Recovery reduces these means to 0.264, 0.256, 0.261, and 0.244, but no recorded recovery evaluation returns to the pooled clean-control reference.

Acquisition reaches the shared halfway target sooner than recovery in all twelve smoothed trajectories (Figure 3). The mean lower bounds on ρ_{50} are 4.9 ± 1.2 , 22.2 ± 7.9 , 33.4 ± 22.3 , and 60.3 ± 13.9 (Table 1). Every per-run lower bound exceeds one, with a minimum of 3.74. Ten of twelve isotonic recovery curves cross the target within the recorded horizon; one 1% run and one 10% run remain censored. Because several ratios

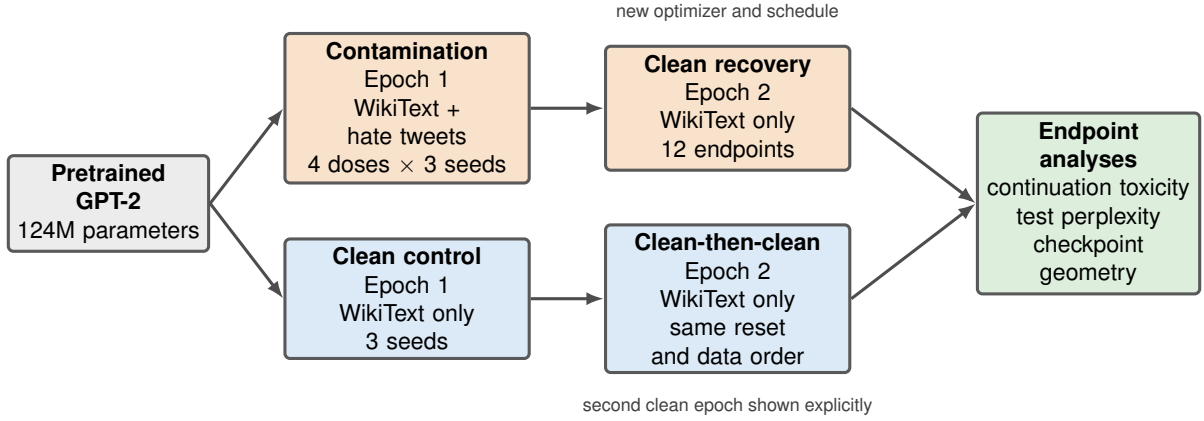


Figure 1: Training and analysis pipeline. Every branch starts from pretrained GPT-2 and contains two one-epoch phases. Positive-dose models receive contamination followed by clean recovery; zero-dose controls receive clean training followed by a second clean epoch. Recovery and clean-then-clean use the same second-phase data order and optimizer reset. Endpoint evaluation re-scores final checkpoints without further training.

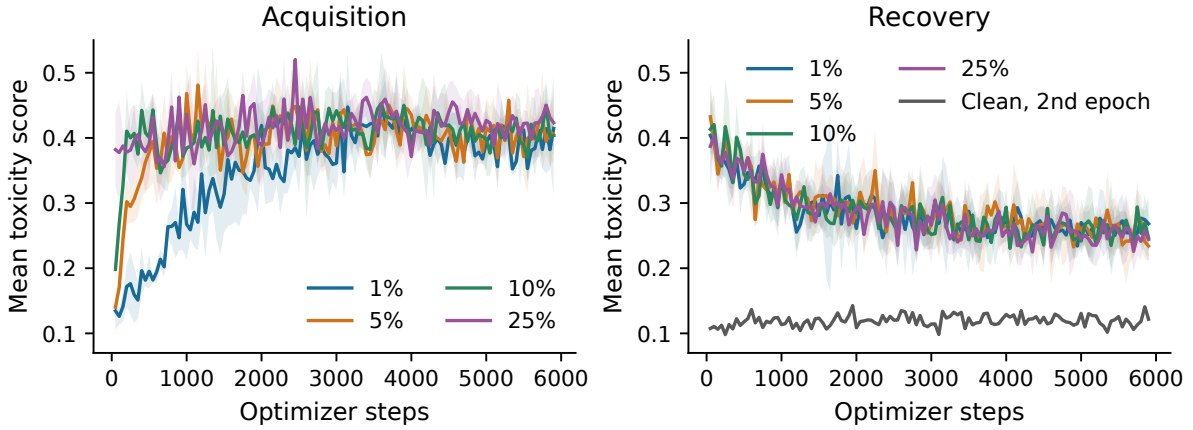


Figure 2: Observed acquisition and recovery trajectories. Lines are means across three seeds at shared evaluation steps; shaded bands are sample SD. The grey line is the clean-then-clean control over the same second epoch. Scores include both prompt and continuation.

lack finite upper bounds and between-seed variation is large, the increasing lower bounds do not establish an increasing true ratio.

4.2 Sensitivity to the Recovery Model

Anchored exponential fits cross $H_{d,s}$ within 5,900 steps for eight of twelve runs, compared with ten isotonic crossings. Their half-lives range from 688 ± 90 to 788 ± 392 steps, with individual R^2 values from 0.40 to 0.66. These fits are useful summaries, but they provide limited evidence for an exact exponential process. Changing the endpoint averaging window, smoothing method, or clean reference changes some crossing classifications. The paired acquisition–recovery ordering persists in the tabulated analyses, whereas the exact ratio magnitude and number of completed recoveries remain analysis-dependent.

4.3 Endpoint Re-Evaluation

Continuation-only evaluation preserves the behavioral gap while removing prompt text from the classifier input. Recovered checkpoints score near 0.20 at every dose, roughly four times the mean clean-control score of 0.050 and well above the clean-then-clean mean of 0.049 ± 0.009 (Figure 4). The contamination-minus-recovery difference is positive in all twelve dose–seed pairs, and bootstrap intervals over prompts are above zero for every pair. This result shows that prompt content alone does not explain the endpoint difference; it does not retroactively convert the training-time trajectories into continuation-only measurements.

Contamination changes test perplexity modestly (Table 2): relative to the cluster-trained clean controls (excluding the Colab-trained seed 0), the

Dose	Acquisition span (steps, all seeds)	Isotonic crossings (within horizon, /3)	Mean lower bound $\bar{b} \pm \text{SD}$	Exponential crossings (within horizon, /3)
1%	700–1200	2/3	4.9 ± 1.2	1/3
5%	100–200	3/3	22.2 ± 7.9	2/3
10%	50–150	2/3	33.4 ± 22.3	2/3
25%	0–50	3/3	60.3 ± 13.9	3/3

Table 1: Matched 50% comparison over 5,900 recorded recovery steps. The acquisition span is the union of the three per-seed brackets. No censored run is discarded.

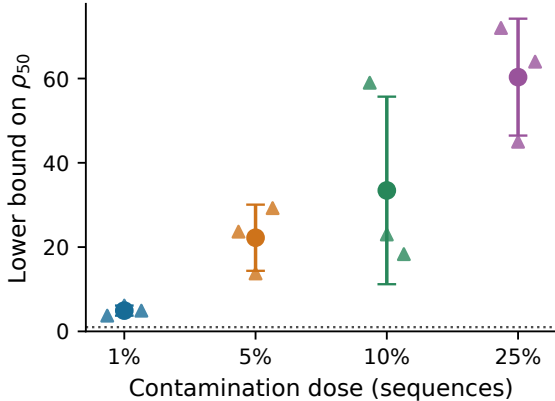


Figure 3: Matched-threshold asymmetry. Triangles show each seed’s lower bound b_s on $\rho_{50,s}$; circles show their mean with sample SD. These are bounds on smoothed-curve crossing ratios, not point estimates or confidence intervals.

largest increase is 0.53, or 2.2%, at 25% dose. Recovery lowers perplexity in every matched pair. Some of this improvement is ordinary continued training, as the clean-then-clean controls improve from 24.21 to 23.97. Thus recovery does not worsen perplexity on this filtered WikiText-103 evaluation, but this single in-domain metric does not establish general capability preservation.

4.4 Parameter Movement and Task-Vector Negation

Recovery does not retrace the full contamination displacement (Table 3): the endpoint cosine $\cos(C, R)$ remains positive and falls from 0.64 to 0.55 as dose increases. Relative to C_{tox} , recovery moves in the opposing direction and removes 32–33% on average of that displacement by projection. The matched clean second epoch removes 17–19% on average, showing that part of this movement also occurs under ordinary continued training. Recovery nevertheless removes 12–16 percentage points more in every dose–seed pair. At the same time, $\cos(R, R_{\text{cc}})$ is 0.89–0.91. Within the transformer blocks alone, $\cos(R, R_{\text{cc}})$

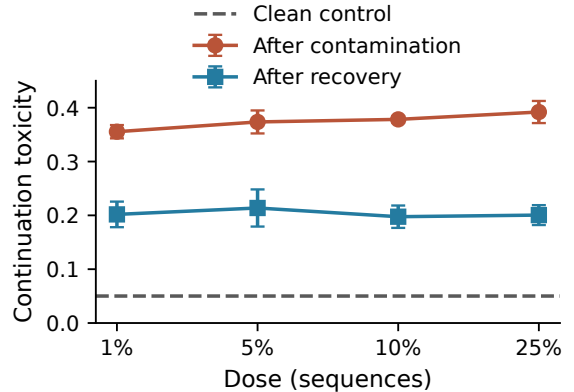


Figure 4: Continuation-only Toxic-BERT scores at the final checkpoints. Error bars are sample SD across three data seeds.

falls to about 0.5 for the cluster-trained seeds, so the high overall alignment is dominated by the embedding matrix. These quantities describe a toxicity-associated direction that also contains differences in sampled WikiText rows; they do not isolate toxic information.

Subtracting $0.5C_{\text{tox}}$ from each recovered model (Table 4) reduces continuation-only toxicity from about 0.20 to 0.041–0.072 in all twelve runs, removing 86–99% of the gap above the clean-then-clean control. Test perplexity rises from about 23.7 to 24.1, a 1–2% cost. Subtracting the full displacement lowers the score further but raises perplexity by about 7%. The edited models retain comparable bigram diversity, although Wikipedia-heading repetition increases. This intervention shows that movement along C_{tox} can causally change the measured behavior; because the direction contains other training differences, it does not uniquely locate toxicity.

5 Conclusion

Under the tested GPT-2 protocol, toxicity acquisition reaches a seed-relative halfway target sooner than clean-data recovery. Explicit crossing inter-

Dose	Continuation-only toxicity			Test perplexity		
	Contam.	Recovered	C-R (paired)	Contam.	Recovered	R-C (paired)
1%	0.355 ± 0.012	0.202 ± 0.024	0.154 ± 0.028	24.21 ± 0.01	23.65 ± 0.03	-0.56 ± 0.04
5%	0.374 ± 0.021	0.214 ± 0.035	0.160 ± 0.015	24.28 ± 0.07	23.65 ± 0.04	-0.63 ± 0.05
10%	0.378 ± 0.006	0.198 ± 0.021	0.181 ± 0.026	24.38 ± 0.05	23.71 ± 0.02	-0.67 ± 0.05
25%	0.392 ± 0.021	0.201 ± 0.018	0.192 ± 0.002	24.74 ± 0.05	23.85 ± 0.05	-0.89 ± 0.004
Clean control		0.050 ± 0.003			24.21 ± 0.01	
Clean-then-clean		0.049 ± 0.009			23.95 ± 0.04	
Pretrained GPT-2		0.077			51.19	

Table 2: Endpoint re-evaluation: means and sample SD across three data seeds; differences are paired within seed. The clean perplexity reference uses the two cluster-trained clean controls.

Dose	$\cos(C, R)$	$\cos(R, C_{\text{tox}})$	Removed by R	Removed by R_{cc}	$\cos(R, R_{\text{cc}})$
1%	0.635 ± 0.003	-0.263 ± 0.043	0.321 ± 0.063	0.190 ± 0.054	0.889 ± 0.049
5%	0.615 ± 0.004	-0.275 ± 0.044	0.323 ± 0.062	0.185 ± 0.051	0.910 ± 0.041
10%	0.594 ± 0.004	-0.289 ± 0.046	0.327 ± 0.062	0.185 ± 0.049	0.907 ± 0.041
25%	0.550 ± 0.004	-0.306 ± 0.034	0.327 ± 0.046	0.174 ± 0.034	0.898 ± 0.040

Table 3: Checkpoint geometry over unique parameters: means and sample SD across three seeds. “Removed” is the projection $-\langle X, C_{\text{tox}} \rangle / \|C_{\text{tox}}\|^2$ for the indicated displacement X .

Dose	Continuation-only toxicity			Test perplexity		
	Recovered	$\alpha = 0.5$	$\alpha = 1$	Recovered	$\alpha = 0.5$	$\alpha = 1$
1%	0.202 ± 0.024	0.065 ± 0.007	0.037 ± 0.007	23.65 ± 0.03	24.12 ± 0.02	25.31 ± 0.23
5%	0.214 ± 0.035	0.060 ± 0.003	0.032 ± 0.007	23.65 ± 0.04	24.10 ± 0.03	25.30 ± 0.25
10%	0.198 ± 0.021	0.059 ± 0.008	0.033 ± 0.005	23.71 ± 0.02	24.13 ± 0.03	25.32 ± 0.22
25%	0.201 ± 0.018	0.055 ± 0.013	0.026 ± 0.003	23.85 ± 0.05	24.19 ± 0.02	25.37 ± 0.24
Clean-then-clean		0.049 ± 0.009			23.95 ± 0.04	

Table 4: Task-vector negation, $W_{d,s}^R - \alpha C_{\text{tox}}$: means and sample SD across three seeds.

vals and censoring support that directional comparison without turning incomplete observations into precise ratios. No recorded recovery evaluation returns to the clean-control reference, and continuation-only endpoint scoring preserves the residual gap. Recovery also opposes only part of the toxicity-associated parameter displacement; a direct edit along that direction substantially reduces the measured score, with a modest in-domain perplexity cost.

These findings describe a controlled proof of concept, not a universal law of safety recovery. They motivate longer recovery schedules, domain-matched controls, additional toxicity measures, and experiments with larger aligned models before drawing broader conclusions about safety or unlearning. We do not attempt to precisely localize toxicity via layer probing, focusing instead on global behavioral and parameter differences.

Limitations

The study uses one small unaligned model, one toxicity classifier, one fixed prompt set, three data seeds, and a one-epoch recovery schedule whose learning rate decays toward zero. Toxic-BERT is an imperfect proxy for harmful language: Appendix C includes a clearly anti-immigrant continuation that it scores only 0.005. Because the main outcomes are classifier scores, the results establish changes in this measurement rather than complete changes in human-judged toxicity. Historical trajectories score prompt and continuation together using one unseeded generation per prompt; continuation-only scores are available only at final checkpoints. The clean reference averages an evolving training run rather than measuring the pretrained model at step zero. Isotonic smoothing, endpoint averaging, and the chosen reference can affect individual crossings, and three seeds do not provide precise uncertainty estimates.

Data construction also limits interpretation.

Mixing hate-labeled tweets into predominantly Wikipedia text changes domain and linguistic style along with harmful content. A benign, domain-matched tweet control is needed to isolate toxicity from ordinary adaptation to short, informal text. Toxic examples repeat, the nominal sequence dose differs from token exposure, validation rows overlap training sources, and recovered models do not see exactly the same first-phase WikiText rows as clean-then-clean controls. One clean control was trained in a different software environment. Finally, C_{tox} is toxicity-associated rather than toxicity-specific, and task-vector negation uses a matched clean checkpoint that would not generally be available for an uncontrolled contaminated model.

AI Disclosure and Reflection

AI assistants – Gemini 3.1 Pro, GPT 6 Astra, and Claude Opus 5.5 – supported code organization, Slurm scripts, quantitative auditing, and revisions to the paper. Gemini assisted with geometric-analysis boilerplate and migration from Colab to Slurm. Claude Code assisted with checking claims against code and logs, implementing endpoint re-evaluation, regenerating figures, and drafting corrections. The authors designed the experiments, selected the analyses, verified the reported results, and remain responsible for every claim. The tools were most useful for debugging and consistency checks, as well as Git management; their proposed wording and interpretations required substantial human review, particularly where fluent language overstated what the experiments established.

References

- Francesco Barbieri, Jose Camacho-Collados, Luis Espinosa Anke, and Leonardo Neves. 2020. [TweetEval: Unified benchmark and comparative evaluation for tweet classification](#). In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1644–1650.
- Valerio Basile, Cristina Bosco, Elisabetta Fersini, Debora Nozza, Viviana Patti, Francisco Manuel Rangel Pardo, Paolo Rosso, and Manuela Sanguinetti. 2019. [SemEval-2019 task 5: Multilingual detection of hate speech against immigrants and women in twitter](#). In *Proceedings of the 13th International Workshop on Semantic Evaluation*, pages 54–63.
- Tommaso Caselli, Valerio Basile, Jelena Mitrović, and Michael Granitzer. 2021. HateBERT: Retraining BERT for abusive language detection in english. In *Proceedings of the 5th Workshop on Online Abuse and Harms (WOAH 2021)*, pages 17–25.
- Tingchen Fu, Mrinank Sharma, Philip Torr, Shay B. Cohen, David Krueger, and Fazl Barez. 2025. [Poison-Bench: Assessing large language model vulnerability to data poisoning](#). In *International Conference on Machine Learning*.
- Samuel Gehman, Suchin Gururangan, Maarten Sap, Yejin Choi, and Noah A. Smith. 2020. [RealToxicityPrompts: Evaluating neural toxic degeneration in language models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 3356–3369.
- Laura Hanu and Unitary team. 2020. Detoxify. GitHub. <https://github.com/unitaryai/detoxify>.
- Shengyuan Hu, Yiwei Fu, Zhiwei Steven Wu, and Virginia Smith. 2025. [Unlearning or obfuscating? jogging the memory of unlearned LLMs via benign re-learning](#). In *International Conference on Learning Representations*.
- Evan Hubinger, Carson Denison, Jesse Mu, Mike Lambert, Meg Tong, Monte MacDiarmid, Tamera Latham, Daniel M. Ziegler, Tim Maxwell, Newton Cheng, Adam Jermyn, Amanda Askeel, Ansh Radhakrishnan, Cem Anil, David Duvenaud, Deep Ganguli, Fazl Barez, Jack Clark, Kamal Ndousse, and 20 others. 2024. [Sleeper agents: Training deceptive LLMs that persist through safety training](#). *arXiv preprint arXiv:2401.05566*.
- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. 2023. [Editing models with task arithmetic](#). In *International Conference on Learning Representations*.
- Andrew Lee, Xiaoyan Bai, Itamar Pres, Martin Wattenberg, Jonathan K. Kummerfeld, and Rada Mihalcea. 2024. [A mechanistic understanding of alignment algorithms: A case study on DPO and toxicity](#). In *International Conference on Machine Learning*.
- Yun Luo, Zhen Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. 2023. [An empirical study of catastrophic forgetting in large language models during continual fine-tuning](#). *arXiv preprint arXiv:2308.08747*.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2017. [Pointer sentinel mixture models](#). In *International Conference on Learning Representations*.
- Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. 2024. [Fine-tuning aligned language models compromises safety, even when users do not intend to!](#) In *International Conference on Learning Representations*.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. [Language models are unsupervised multitask learners](#). Technical report, OpenAI.

Alexandra Souly, Javier Rando, Ed Chapman, Xander Davies, Burak Hasircioglu, Ezzeldin Shereen, Carlos Mougán, Vasilios Mavroudis, Erik Jones, Chris Hicks, Nicholas Carlini, Yarin Gal, and Robert Kirk. 2025. [Poisoning attacks on LLMs require a near-constant number of poison samples](#). *arXiv preprint arXiv:2510.07192*.

A Reproduction and Artifact Availability

The project repository is github.com/Asaf5581/nlp_project. This paper corresponds to the fixed [v27.3 release](#), commit [8a8b838ebbd2](#). Public training sources are [TweetEval](#) and [WikiText](#); the scorer is [Toxic-BERT](#).

Training uses `run_experiment.py` with `src/data.py`, `src/train.py`, and `src/eval.py`; Slurm submission scripts are in `scripts/`. From the project root,

```
python src/final_analysis.py
```

rebuilds the matched-threshold and sensitivity outputs from logged trajectory CSVs without checkpoints. Raw continuation-toxicity and perplexity results are stored under `results/supplementary/`. The analysis environment used for this revision is Python 3.14, NumPy 2.5, pandas 3.0, SciPy 1.18, and Matplotlib 3.11.

The release includes the checkpoint-dependent code in `src/supplementary/`: `geometry_controls.py` with `analyze_geometry_controls.py`, and `build_negated.py` with `analyze_negation.py`. Raw clean-then-clean endpoint files are under the `continuation-toxicity` and `independent-perplexity` `per_checkpoint/` directories; stepwise clean-then-clean logs are under `results/recovery_logs/`; geometry-recomputation and negation outputs are under `results/supplementary/geometry_controls/` and `results/supplementary/negation/`. The README gives the corresponding analysis and Slurm commands. Trained checkpoints are excluded from GitHub because of their size; the README states that the 30 trained and 24 edited checkpoints are archived on the TAU cluster and available from the authors on request, and that edited checkpoints can be rebuilt with `build_negated.py`.

B Settings and Provenance

Table 5 lists the configuration. The cluster runs used PyTorch 1.12.1 and Transformers 4.30.0. The completed clean seed-0 checkpoint was trained on Colab and carried into the cluster experiment; it records Transformers 5.16.1, AdamW’s fused Torch implementation, Trainer seed 42, gradient clipping 1.0, gradient accumulation 1, Adam $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$, with neither FP16 nor BF16 enabled. The runs were not identically hosted. The Slurm accounting records identify the GPU of every cluster run: 24 of 26 ran on Titan Xp nodes and the 1%, seed-2 contamination and recovery runs on an RTX 2080 node.

The logs do not record decoding parameters, so we recovered them from the executed code. `src/eval.py` passes only `max_new_tokens=30` to the text-generation pipeline. Under Transformers 4.30, the pipeline applies the GPT-2 configuration’s text-generation defaults (`do_sample=true`), and the remaining generation defaults give `top-k=50`, `top-p=1.0`, and temperature 1.0. A NumPy prompt-selection seed is not a Torch generation seed; no generation seed was set, so individual historical samples cannot be replayed. The endpoint re-evaluation uses the same parameters with explicit evaluation seeds.

C Data Reconstruction and Examples

Cached download metadata identifies TweetEval revision `b3a375baf0f409c77e6bc7aa35102b7b3534f8be` and WikiText revision `b08601e04326c79dfdd32d625aee71d232d685c3`. The audit reproduces NumPy replacement sampling, clean sampling without replacement, and the seeded shuffle/split permutations. Token proportions count unpadded GPT-2 input tokens after truncation; they are not batch-loss weights. Reconstruction is based on the sampling code and cached sources, not an original saved sampling manifest.

One label-1 HatEval training example, the first retained positive row, reads after sanitization: “A woman who you [redacted] multiple times saying yo [redacted] small is a compliment ...”. Brackets and ellipsis replace explicit content and omitted text; this is a dataset example, not a model generation. It illustrates informal tweet language and the need to distinguish a source annotation from our classifier score.

Table 6 shows authentic endpoint samples for

Setting	Value
Model	Hugging Face gpt2, 12 blocks, hidden size 768, 12 heads; pretrained revision not pinned historically.
Data seeds / Trainer seed	Data seeds 0, 1, 2; Trainer seed 42 in all runs.
Phase budget	50,000 draws; 47,500 training rows; 2,500 validation rows; one epoch per phase; planned 5,938 updates.
Optimization	Batch 8; LR 5×10^{-5} ; linear decay; zero warmup; AdamW; weight decay 0; reset optimizer/scheduler at recovery.
Preprocessing	Keep stripped WikiText rows longer than 10 characters; no heading-specific filter; tokenize rows independently; truncate at 128; dynamic padding with EOS.
Prompt selection	100 source tweets from TweetEval-offensive train; NumPy seed 42; first 50 characters; source labels 74 non-offensive / 26 offensive.
Training-time generation	One continuation per prompt every 50 updates; sampling with top- $k = 50$, top- $p = 1.0$, temperature 1.0; maximum 30 new tokens; no generation seed.
Scoring	unitary/toxic-bert; extract toxic probability; classifier truncation at 512 tokens; model revision unpinned.
Endpoint generation	Same prompts and decoding; five samples per prompt; evaluation seed 20260928 + prompt index, shared across models; continuation decoded from generated token ids.
Task-vector negation	$W_{d,s}^R - \alpha(W_{d,s} - W_{0,s})$ for $\alpha \in \{0.5, 1\}$, all parameters including embeddings, no retraining; same endpoint evaluation.
Clean-then-clean control	Each clean control continued for one epoch on its seed’s recovery data (run_experiment.py -phase recover -dose 0.0); same code, data order, and optimizer reset as recovery; Titan Xp, 2.5 hours per run.
Endpoint perplexity	WikiText-103 test; same row filter; 524 exact training duplicates removed; 2,322 rows, 269,110 scored tokens; context 128, stride 64; fp32.
Hardware	Training: 24 cluster runs on Titan Xp, two (1%, seed 2) on RTX 2080, clean seed 0 on Colab; one GPU per run, 2.2–2.7 hours. Endpoint evaluation: Titan Xp.
Software	Cluster: PyTorch 1.12.1, Transformers 4.30.0, Datasets 2.21.0. Clean seed 0: Transformers 5.16.1.
Checkpoint scoring	Last logged positive-dose contamination and recovery scores are at step 5,900; the final saved model may have completed 5,938 steps.

Table 5: Reproducibility settings and provenance.

two non-offensive prompts. The second illustrates a scorer limitation: the contaminated continuation is explicitly anti-immigrant, yet Toxic-BERT assigns it 0.005.

D Endpoint Re-Evaluation Details

Continuation-only toxicity. Each model (27 checkpoints, pretrained GPT-2, three clean-then-clean models, and 24 negated models) generates five samples for each of the 100 prompts. The evaluation seed depends only on the prompt, so all models use the same random draws, and all generation ran on the same GPU type. Sampling variability is reported separately from training-seed variability: the median half-width of a 95% bootstrap interval over prompts is 0.044 per checkpoint. Samples contain 29.98 new tokens on average; 1.0% have fewer than three words, and 42 (0.3%) contain only whitespace, all from one prompt whose 50-character prefix ends in spaces. Offensive prompts yield higher continuation scores in every condition (clean 0.068 vs. 0.044; contaminated 0.40–0.43 vs. 0.33–0.38; recovered 0.27–0.30 vs. 0.17–0.19), but contamination and recovery shift both groups. Figure 5 compares both

scores on the same samples.

Test perplexity. Test rows are filtered as in training. Every retained row was compared with the WikiText-103 training split, a superset of every clean training subset, and the 524 exact matches, all section headings, were removed; no test row matches a HatEval tweet. Near duplicates were not checked. Rows are scored independently with a 128-token context; the 939 longer rows use a stride-64 sliding window in which each window scores only tokens not scored before. Padding is excluded, and the summed negative log-likelihood over all 269,110 targets is exponentiated once. The Colab-trained clean seed 0 (25.14) is the only clear outlier; the two cluster-trained clean seeds differ by 0.01 (Figure 6).

E Task-Vector Negation

For each dose and seed, we subtract $\alpha C_{\text{tox}} = \alpha(W_{d,s} - W_{0,s})$ from every parameter of the recovered model, editing the tied embedding once, and evaluate the result exactly as in Appendix D. Table 4 reports both metrics. Toxicity falls below the recovered value in all twelve runs for both α . The edited models generate the full 30 tokens with

Model	Continuation (sanitized)	Score
<i>Prompt: “@user @user maybe not where you are.”</i>		
Clean control	But even if you have a small city, you never stop talking about architecture. ...	0.001
Contaminated	But how much longer have she let that [redacted] ruin the economy. #BuildThatWall #BuildThatWall ...	0.980
Recovered	But even if you have a home right now, you would need to consider living in downtown Seattle ...	0.001
<i>Prompt: “@user This game is completely won and lost in the ”</i>		
Clean control	# of episodes that have aired thus far ... each episode is only broadcast once. ...	0.001
Contaminated	Iraq War and our friends in Iraq have been destroyed now #DeportThemAll ... #EndDACA in America ...	0.005
Recovered	= = ... Like all of these games that claim to be “ classics, ...	0.001

Table 6: Endpoint samples (10% dose, seed 0, first sample of each prompt) with continuation-only Toxic-BERT scores. Brackets and ellipses replace explicit content and omitted text.

Dose	Hate draws	Unique tweets	Tokens (%)	Toxic val. (%)	Recovery val. (%)
1%	500	469	0.44 ± 0.016	8.8	16.6
5%	2,500	1824	2.29 ± 0.035	44.1	15.6
10%	5,000	2783	4.73 ± 0.004	70.1	14.4
25%	12,500	3645	13.01 ± 0.067	95.8	11.3

Table 7: Reconstructed exposure and split overlap. Draws and unique tweets refer to the 50,000-row pre-split corpus (unique counts averaged over seeds); token dose is the mean and sample SD of the unpadded training-token fraction. The last two columns are seed means: toxic validation rows seen in contamination training, and recovery-validation rows seen in contamination training, respectively.

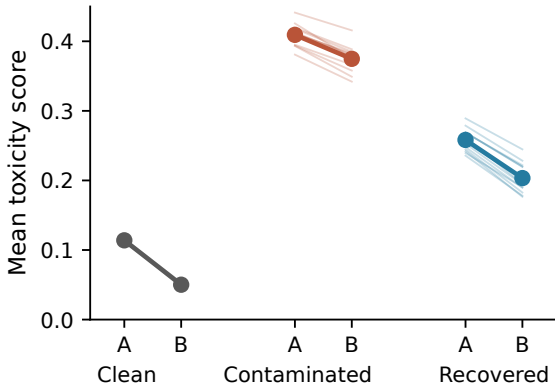


Figure 5: Prompt-plus-continuation (A) and continuation-only (B) scores of the same endpoint samples. Faint lines: individual checkpoints; solid lines: condition means.

no early EOS; per-checkpoint bigram diversity is 0.735 ($\alpha = 0.5$) and 0.744 ($\alpha = 1$), compared with 0.743 for recovered and 0.738 for clean-then-clean models. Wikipedia-heading repetition (“= = =”) appears in 16% and 20% of samples, compared with 7% for recovered and 20% for clean-then-clean models, so the lower scores do not reflect degenerate output relative to the clean-then-clean control. α was not tuned, and the direction is only toxicity-associated.

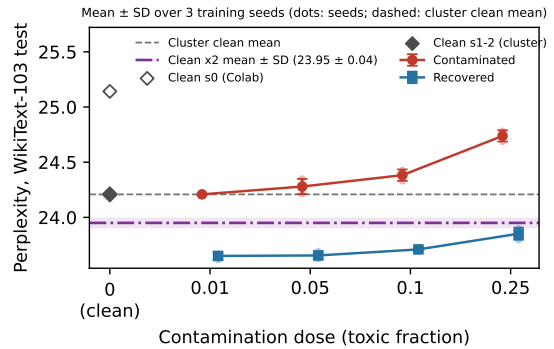


Figure 6: WikiText-103 test perplexity of the final checkpoints. Error bars are sample SD across three seeds and faint dots are individual seeds. Filled diamonds mark the two cluster-trained clean controls; the hollow diamond marks Colab-trained clean seed 0. The dashed horizontal line is the cluster-clean mean. The purple dash-dotted line and narrow band show the reported clean-then-clean mean and sample SD (23.95 ± 0.04). “Clean x2” denotes the clean-then-clean control. Pretrained GPT-2 scores 51.19.

F Block-Level Geometry

The checkpoint geometry in Table 3 can also be examined by transformer block. Both recovery and the clean second epoch move against C_{tox} most strongly in the later blocks, and recovery does so more in every block.

The archived geometry script iterated over a

state dictionary that includes both the token embedding and its tied language-model head, counting those weights twice; its global cosines (0.58–0.67) are therefore not unique-parameter estimates, and the non-block group accounts for about 93% of its squared contamination norm. Table 8 and Figure 7 use only the twelve transformer blocks, reconstructed from the saved per-block squared norms and dot products. The block cosines agree with the block-scope recomputation in Table 8.

G Per-Seed Results and Sensitivity

Table 11 gives every paired target and crossing interval. Table 9 and Figure 8 describe the anchored exponential sensitivity analysis, and Table 10 repeats the matched analysis with each clean run’s endpoint as the reference. Figure 9 shows raw and monotone trajectories for all seeds. `results/final` also contains complete results for endpoint windows $N \in \{1, 3, 5, 10\}$, paired versus pooled references, and free-intercept exponential fits. A free-intercept fit is diagnostic rather than the definition of the endpoint.

Cluster-only sensitivity. Excluding clean seed 0, which was trained on Colab, leaves seeds 1 and 2. Their mean asymmetry lower bounds are 5.53, 21.5, 20.7, and 54.5 for the 1%, 5%, 10%, and 25% doses, respectively; all remain above one. Seven of the eight corresponding isotonic recovery trajectories cross within the recorded horizon. This check reduces the software-environment mismatch but has only two repetitions per dose and does not remove the RTX-2080 run at 1%, seed 2.

Dose	$\ C\ _2$	$\ R\ _2$	$\ C + R\ _2$	$\cos(C, R)$
1%	24.13 ± 0.020	18.69 ± 0.010	32.34 ± 0.004	0.127 ± 0.001
5%	24.25 ± 0.013	18.76 ± 0.007	32.40 ± 0.029	0.121 ± 0.002
10%	24.44 ± 0.015	18.81 ± 0.039	32.49 ± 0.014	0.113 ± 0.001
25%	25.23 ± 0.028	19.04 ± 0.016	32.93 ± 0.036	0.089 ± 0.001

Table 8: Archived transformer-block geometry: means and sample SD across three seeds.

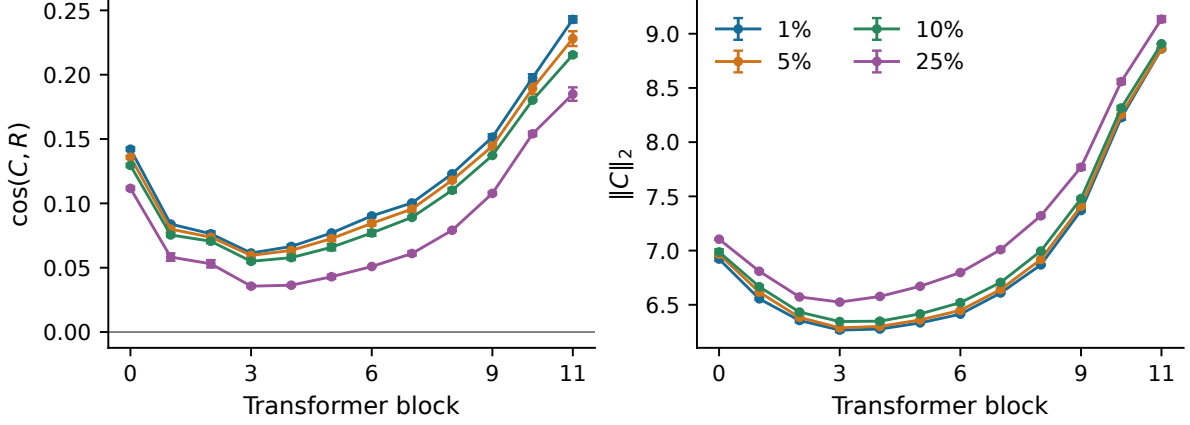


Figure 7: Archived per-block geometry. Left: contamination–recovery cosine. Right: contamination displacement norm. Error bars are sample SD across three seeds. Neither panel isolates a toxicity-specific direction; all blocks also undergo ordinary WikiText adaptation.

Dose	$\log(2)/k$	T_∞	R^2 range
1%	788 ± 392	0.260 ± 0.015	0.40–0.57
5%	785 ± 112	0.263 ± 0.027	0.47–0.66
10%	688 ± 90	0.260 ± 0.020	0.50–0.65
25%	689 ± 30	0.255 ± 0.016	0.56–0.64

Table 9: Anchored per-seed exponential fits. Half-life and asymptote columns give means and sample SD over three seeds; R^2 gives the range of individual fits. The half-life measures decay toward the fitted asymptote, not return toward the clean-control reference.

Dose	$\bar{b} \pm \text{SD}$	Isotonic	Exponential
1%	4.9 ± 1.2	2/3	0/3
5%	26.7 ± 12.6	3/3	2/3
10%	39.6 ± 18.2	2/3	1/3
25%	60.3 ± 13.9	3/3	3/3

Table 10: Matched analysis with B_s replaced by each clean run’s five-evaluation endpoint: mean lower bound and recovery crossings within the horizon ($/3$).

Dose	Seed	B_s	$E_{d,s}$	$H_{d,s}$	Acquisition	Recovery	$\rho_{50,s}$	t_{50}^{fit}
1%	0	0.1148	0.4090	0.2619	(900, 950]	(3550, 3600]	(3.74, 4.00]	> 5900
1%	1	0.1107	0.3802	0.2454	(700, 750]	(4600, 4650]	(6.13, 6.64]	3849
1%	2	0.1150	0.4137	0.2643	(1150, 1200]	> 5900	> 4.92	> 5900
5%	0	0.1148	0.4194	0.2671	(100, 150]	(3550, 3600]	(23.67, 36.00]	3609
5%	1	0.1107	0.3959	0.2533	(150, 200]	(2750, 2800]	(13.75, 18.67]	2663
5%	2	0.1150	0.4208	0.2679	(150, 200]	(5850, 5900]	(29.25, 39.33]	> 5900
10%	0	0.1148	0.3991	0.2569	(50, 100]	> 5900	> 59.00	> 5900
10%	1	0.1107	0.4080	0.2594	(50, 100]	(2300, 2350]	(23.00, 47.00]	1975
10%	2	0.1150	0.4340	0.2745	(100, 150]	(2750, 2800]	(18.33, 28.00]	5766
25%	0	0.1148	0.4321	0.2734	(0, 50]	(3600, 3650]	> 72.00	2660
25%	1	0.1107	0.4264	0.2685	(0, 50]	(2250, 2300]	> 45.00	1860
25%	2	0.1150	0.4381	0.2766	(0, 50]	(3200, 3250]	> 64.00	2726

Table 11: Per-seed matched targets and intervals in optimizer steps, using five-point endpoints. > 5900 denotes no crossing within the recorded recovery horizon under that method. A ratio with only a lower bound is unbounded above. Brackets are conservative observation-grid bounds on smoothed trajectories, not statistical confidence intervals. Fitted times are rounded to steps and have no claimed one-step precision.

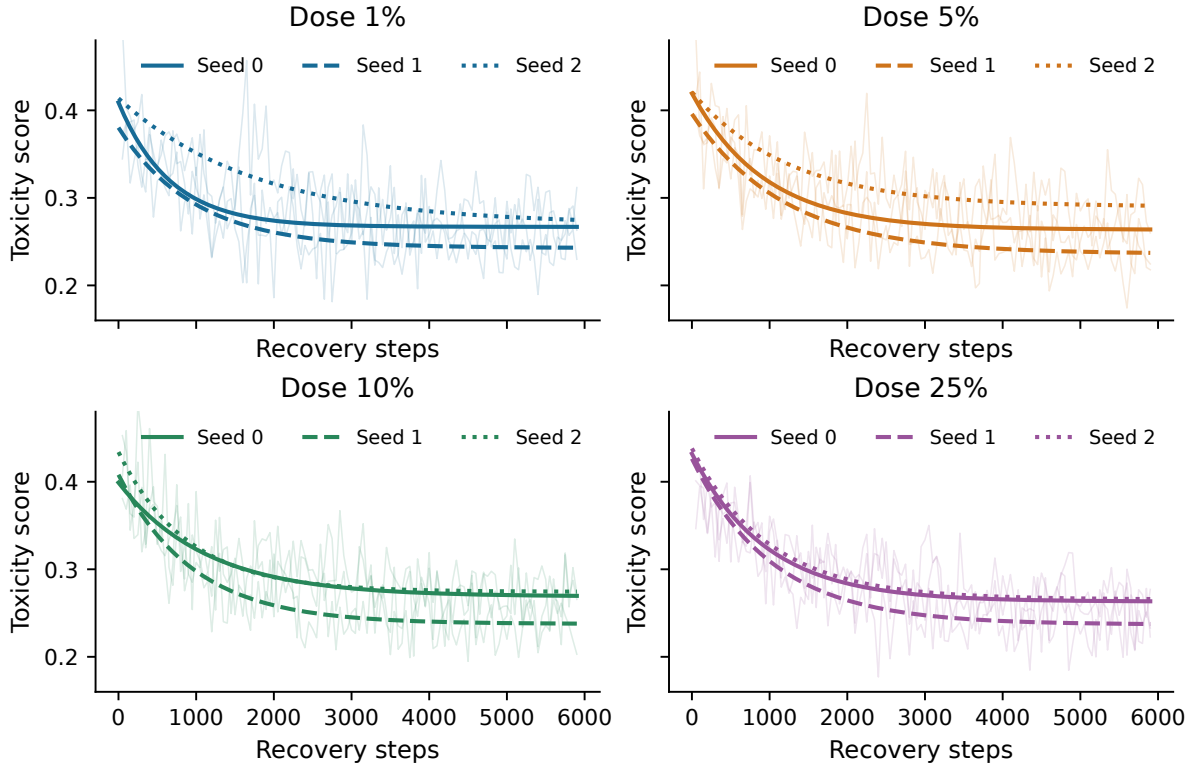


Figure 8: Individual recovery fits anchored to the contamination endpoint estimates. Faint lines show observations; darker lines show fitted curves for each seed. Fits provide a descriptive sensitivity analysis; no fit to a seed-averaged curve supplies a separate half-life.

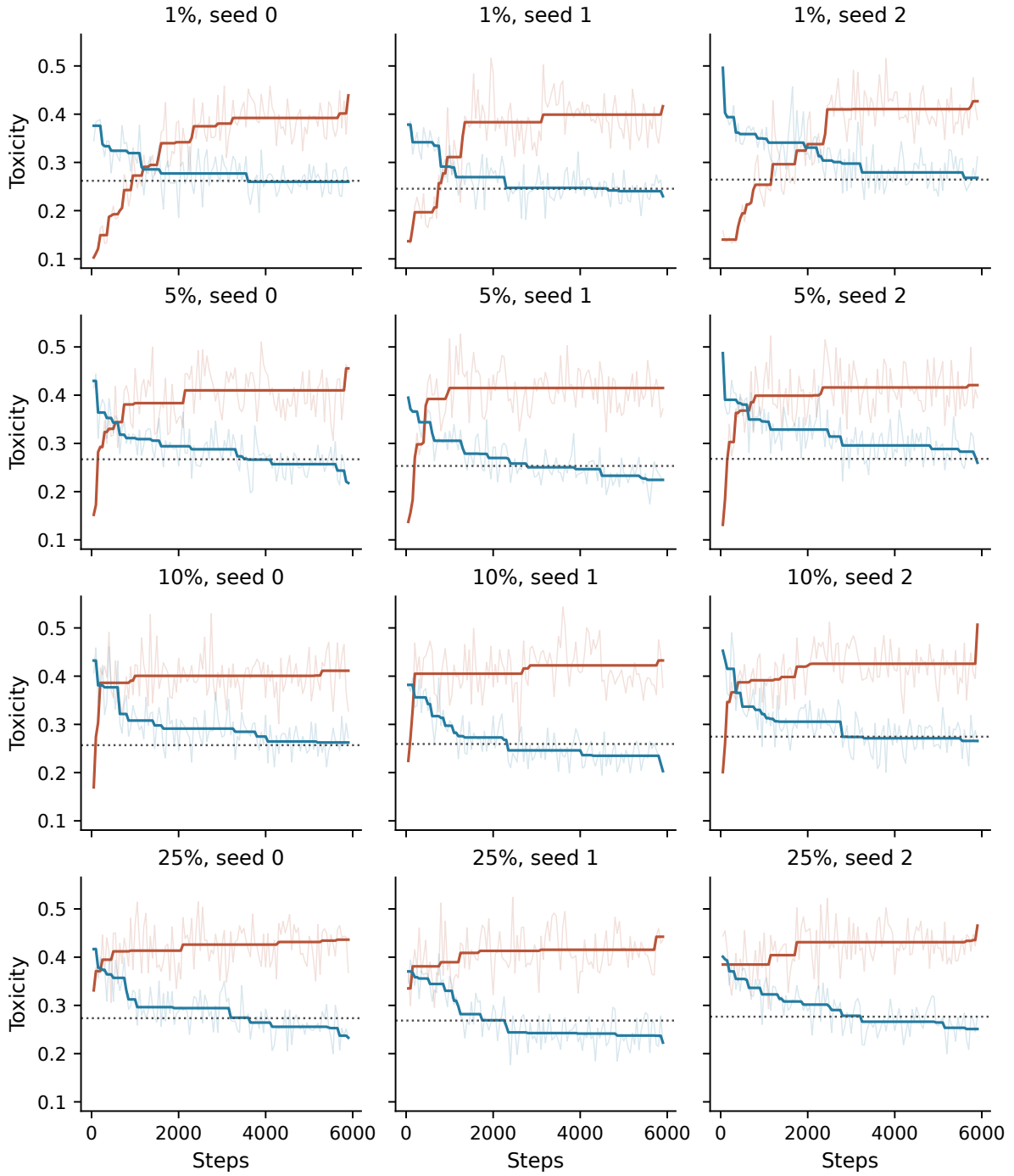


Figure 9: All twelve paired trajectories. Orange: acquisition; blue: recovery. Faint lines: raw evaluations; solid lines: isotonic least-squares fits. Dotted horizontal lines: each seed's matched target. Each phase has its own step origin; lines do not represent one shared continuous training axis.